

REFATORAÇÃO DE CODE SMELLS EM PROJETO OPEN-SOURCE

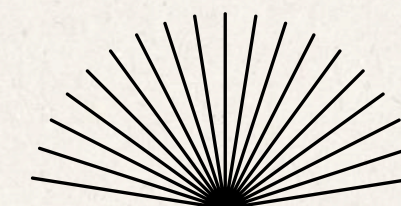
Refatoração guiada por LLM no projeto open-source: **SUPERVISION**

PRESENTED BY:

Julian Cardoso e Edivar Cruz

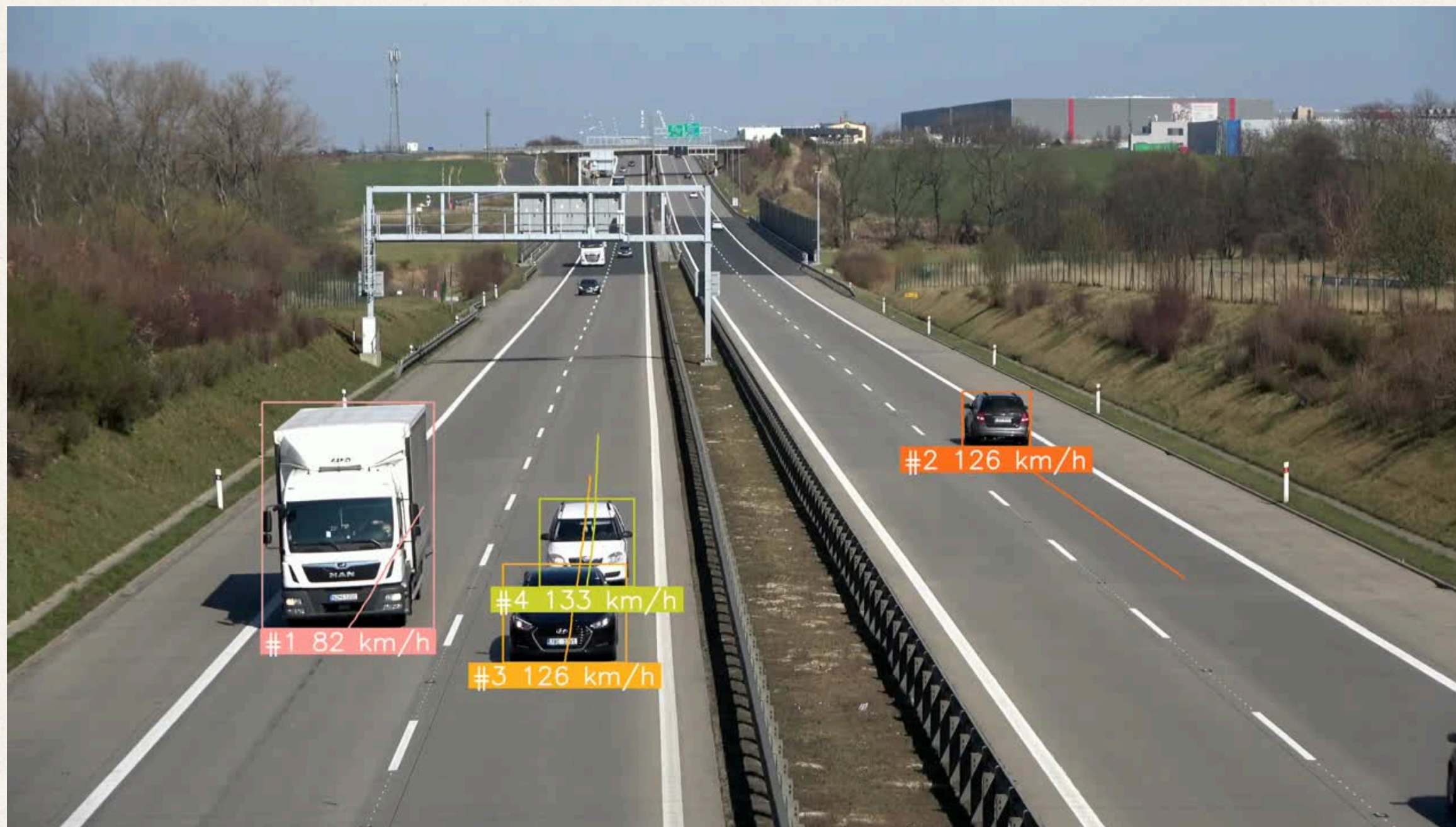
PRESENTED TO:

Prof. Carla Illane



Projeto analisado: Supervision

Biblioteca Python para
visão computacional
Utilizada para
processamento e anotação
de detecções



Projeto analisado:



02/15

Foram encontrados 33 smells:

- 15 too-many-instance-attributes
- 7 too-many-statements
- 11 too-many-branches

Modelos usados:

- DeepSeek Flash V4 (too-many-statements)
- GPT 5.1 Mini

Antes da Refatoração

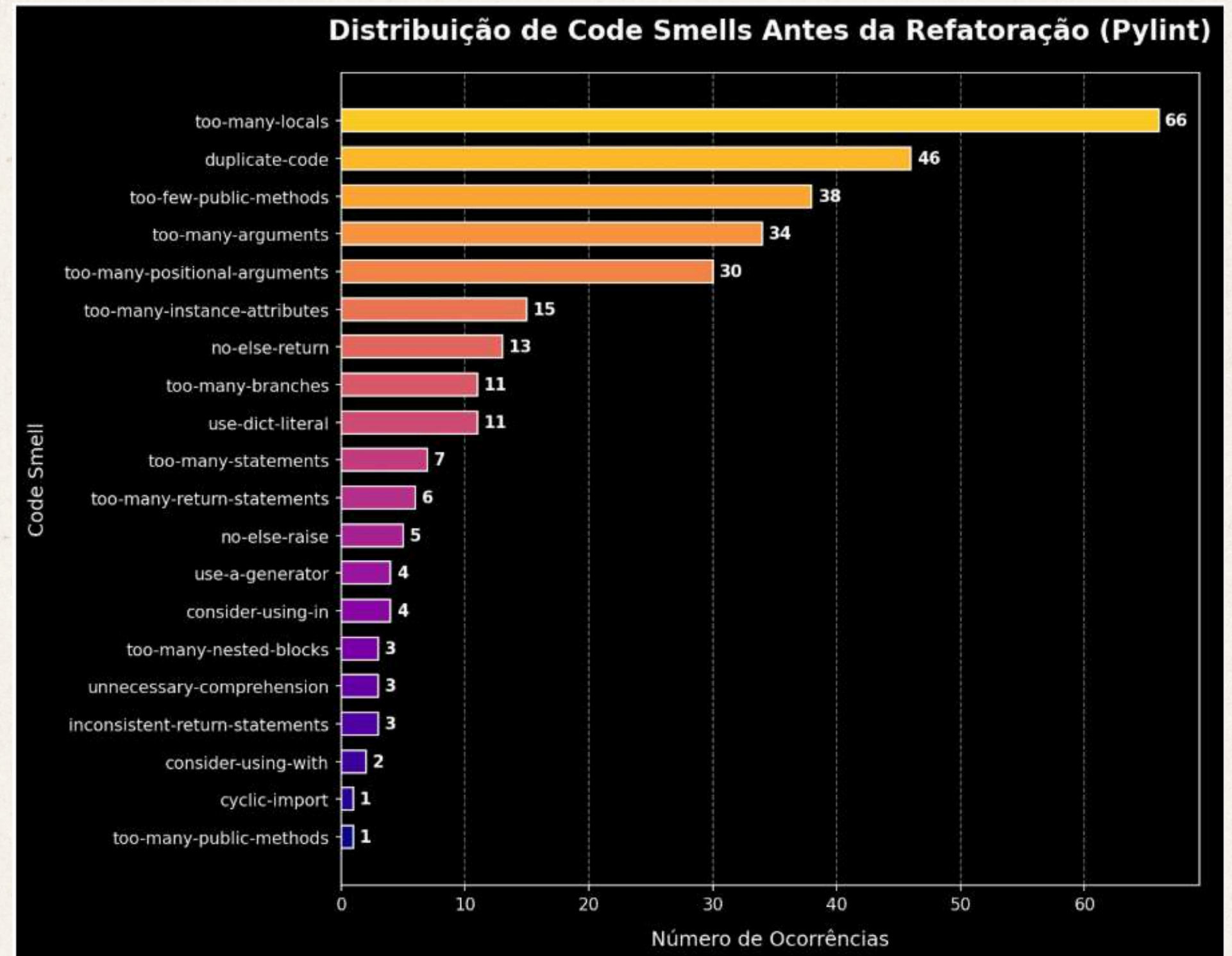
03/15

Qualidade PyLint: 7.98/10

Maior CC: 33

Testes que passaram: 2064

Testes que não passaram: 4



Radon - Antes da refatoração

Piores métricas

04/15

arquivo	funções	cc_medi a	cc_ma x	cc_som a	pior_rank	pior_funcao
supervision/detection/utils/internal.py	8	10.12	22	81	D	<code>process_roboflow_result</code>
supervision/detection/vim.py	15	8.67	33	130	E	<code>from_google_gemini_2_0</code>
supervision/detection/utils/ <u>poligon.py</u>	2	8.00	9	16	B	<code>filter_polygons_by_area</code>
supervision/metrics/mean_avarage_precision.py	32	7.03	29	225	D	<code><u>accumulate</u></code>
supervision/tracker/byte_tracker/core.py	7	6.86	26	48	D	<code>update_with_tensors</code>

Melhores métricas

04/15

arquivo	funções	cc_media	cc_max	cc_soma	pior_rank	pior_funcao
supervision/annotators/base.py	1	1.0	1	1	A	annotate
supervision/geometry/core.py	11	1.0	1	11	A	list
supervision/metrics/core.py	3	1.0	1	3	A	update
supervision/tracker/by_tracker/utils.py	4	1.25	2	5	A	init
supervision/tracker/by_tracker/kalman_filter.py	6	1.33	2	8	A	init

Complexidade ciclométrica por função Antes da refatoração

Piores métricas

04/15

arquivo	tipo	nome	rank_c	complexity
src/supervision/key_points/core.py	method	KeyPoints.getitem	E	39
src/supervision/detection/vlm.py	function	from_google_gemini_2_5	E	33
src/supervision/metrics/mean_average_precision.py	method	COCOEvaluator_accumulae	D	29
src/supervision/metrics/mean_average_precision.py	method	COCOEvaluator_evaluate_image	D	28
src/supervision/tracker/byte_tracker/core.py	method	ByteTrack.update_with_tensors	D	26

Melhores métricas

04/15

arquivo	tipo	nome	rank_cc	complexity
src/supervision/annotators/base.py	method	<u>BaseAnnotator.annotate</u>	A	1
src/supervision/annotators/core.py	method	_normalize_color_input	A	1
src/supervision/annotators/core.py	function	<u>BaseLabelAnnotator.init</u>	A	1
src/supervision/annotators/core.py	function	<u>BoxAnnotator.init</u>	A	1
src/supervision/annotators/core.py	function	<u>OrientedBoxAnnotator.init</u>	A	1

Índice de Manutenibilidade Antes da refatoração

Piores métricas

04/15

arquivo	mi	rank_mi
src/supervision/annotators/core.py	0.5164	C
src/supervision/metrics/mean_average_precision.py	11.3722	C
src/supervision/detection/core.py	15.2378	B
src/supervision/detection/compact_mask.py	25.2684	B
src/supervision/detection/utils/converters.py	26.8962	A

Melhores métricas

04/15

arquivo	mi	rank_mi
src/supervision/config.py	100.0	A
src/supervision/annotators/base.py	100.0	A
src/supervision/annotators/init.py	100.0	A
src/supervision/assets/list.py	100.0	A
src/supervision/assets/init.py	100.0	A

Métricas brutas - Antes da refatoração

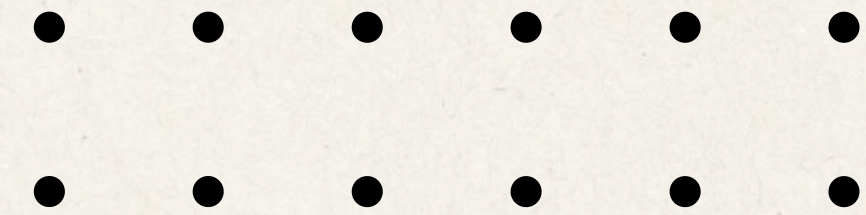
04/15

loc total	lloc total	sloc total	comments total	multi
33953	10762	19554	522	9129

CodeCarbon - Antes da refatoração

duration	emissions	emissions_rate	cpu_energy	ram_energy	energy_consumed
88.1082	1.9513e-05	2.2147e-07	7.0576e-05	1.2783e-04	1.9841e-04

Depois da Refatoração

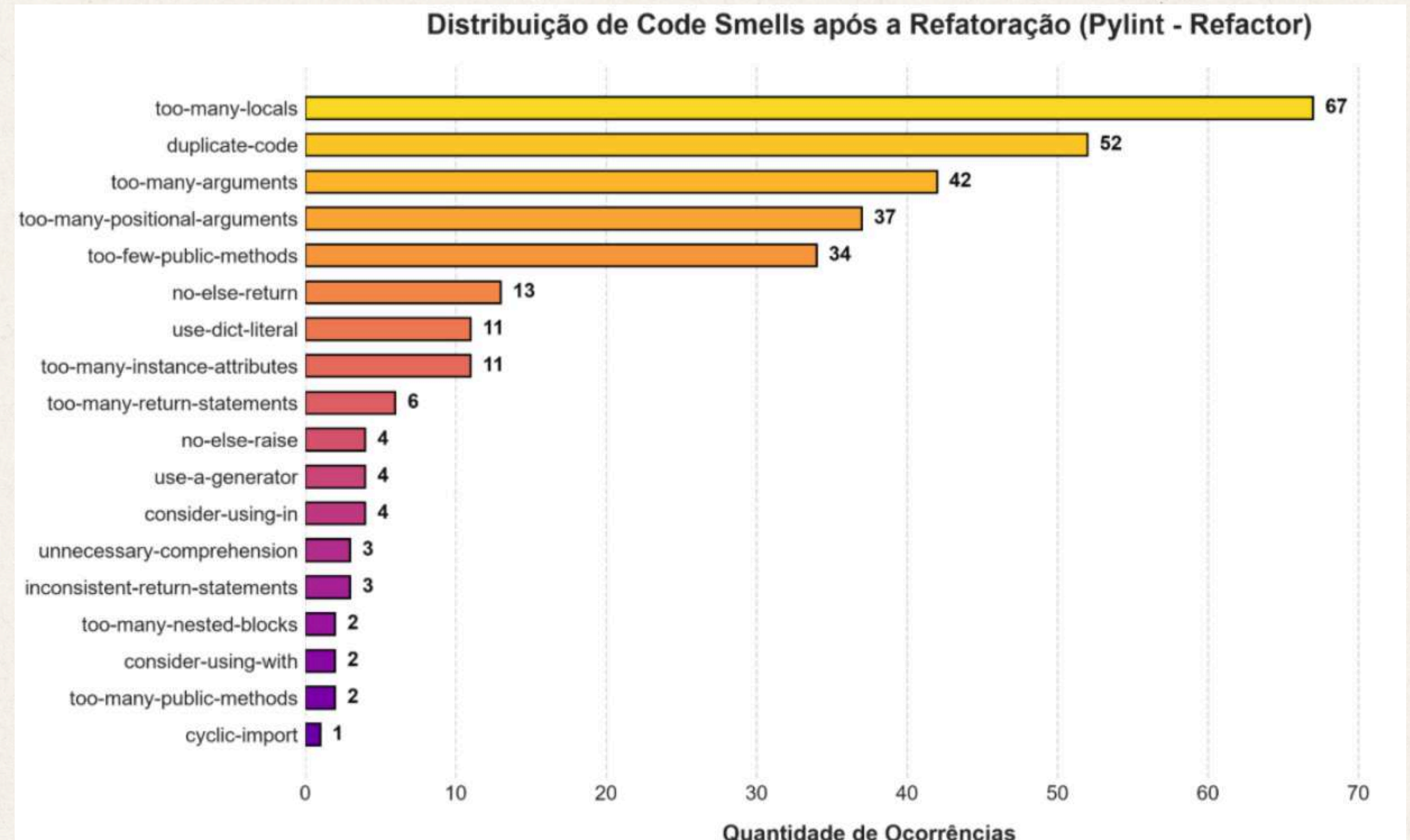


Qualidade PyLint: 7.72/10

Maior CC: 39

Testes que passaram: 2031

Testes que não passaram: 44



Radon - Depois da refatoração

Piores métricas

04/15

arquivo	funções	cc_media	cc_max	cc_soma	pior_rank	pior_funcao
src/supervision/key_points/core.py	22	6.14	39	135	E	getitem
src/supervision/detection/tools/smooth.py	4	6.50	9	26	B	get_track
src/supervision/detection/track_utils/polygons.py	2	8.00	9	16	B	filter_polygons_by_area
src/supervision/detection/track/vlm.py	22	6.05	33	133	C	from_google_gemini_2_5
src/supervision/detection/track_utils/internal.py	8	10.12	22	81	D	process_robotflow_result

Melhores métricas

04/15

arquivo	funções	cc_media	cc_max	cc_soma	pior_rank	pior_funcao
src/supervision/annotators/base.py	1	1.00	1	1	A	annotate
src/supervision/geometry/core.py	11	1.00	1	11	A	list
src/supervision/metrics/core.py	3	1.00	1	3	A	update
src/supervision/tracker/byte_tracker/Utils.py	4	1.25	2	5	A	init
src/supervision/detection/tools/polygon_zone.py	22	1.27	4	28	A	annotate



Complexidade ciclométrica por função Depois da refatoração

Piores métricas

04/15

arquivo	tipo	nome	rank_cc	complexity
src/supervision/tracker/byte_tracker/core.py	method	update_with_tensors	D	26
src/supervision/metrics/mean_average_precision.py	method	<u>evaluate_image</u>	D	28
src/supervision/metrics/mean_average_precision.py	method	<u>accumulate</u>	D	29
src/supervision/detection/vlm.py	function	from_google_gemini_2_5	E	33
src/supervision/key_points/core.py	method	getitem	E	39



Melhores métricas

04/15

arquivo	tipo	nome	rank_cc	complexity
src/supervision/annotators/base.py	method	annotate	A	1
src/supervision/annotators/core.py	function	normalize_color_input	A	1
src/supervision/annotators/core.py	method	init	A	1
src/supervision/annotators/core.py	method	color	A	1
src/supervision/annotators/core.py	method	color_lookup	A	1



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Índice de Manutenibilidade Depois da refatoração

Piores métricas

04/15

arquivo	mi	rank_mi
src/supervision/annotators/core.py	0.0000	C
src/supervision/metrics/mean_average_precision.py	5.4662	C
src/supervision/detection/core.py	15.2378	B
src/supervision/detection/compact_mask.py	25.2684	A
src/supervision/detection/utils/converters.py	26.8962	A

Melhores métricas

04/15

arquivo	mi	rank_mi
src/supervision/metrics/init.py	100.0000	A
src/supervision/metrics/utils/utils.py	100.0000	A
src/supervision/metrics/utils/init.py	100.0000	A
src/supervision/tracker/init.py	100.0000	A
src/supervision/tracker/byte_tracker/init.py	100.0000	A



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Métricas brutas - Depois da refatoração

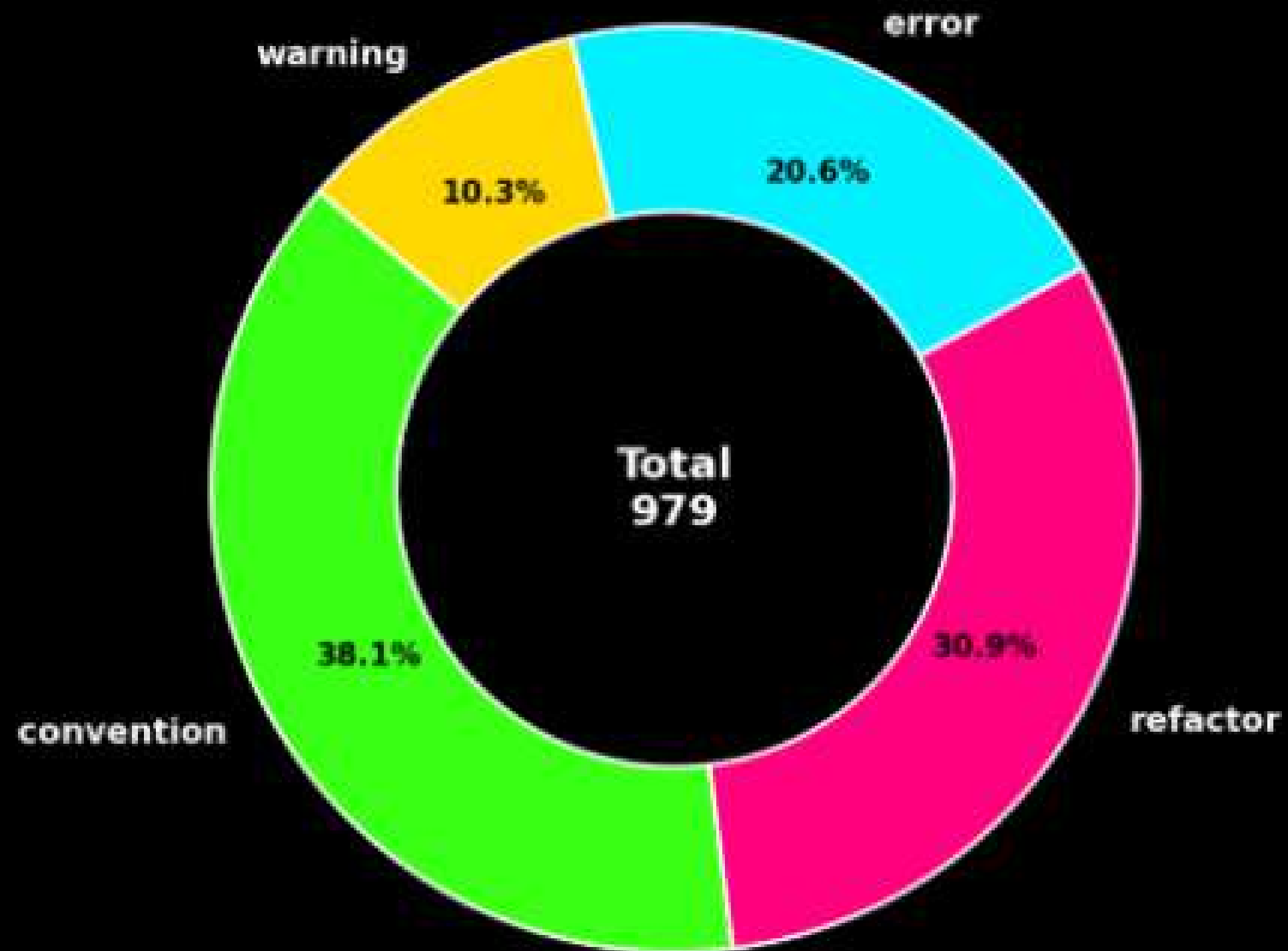
04/15

loc total	lloc total	sloc total	comments total	multi
34876	11386	20295	516	9110

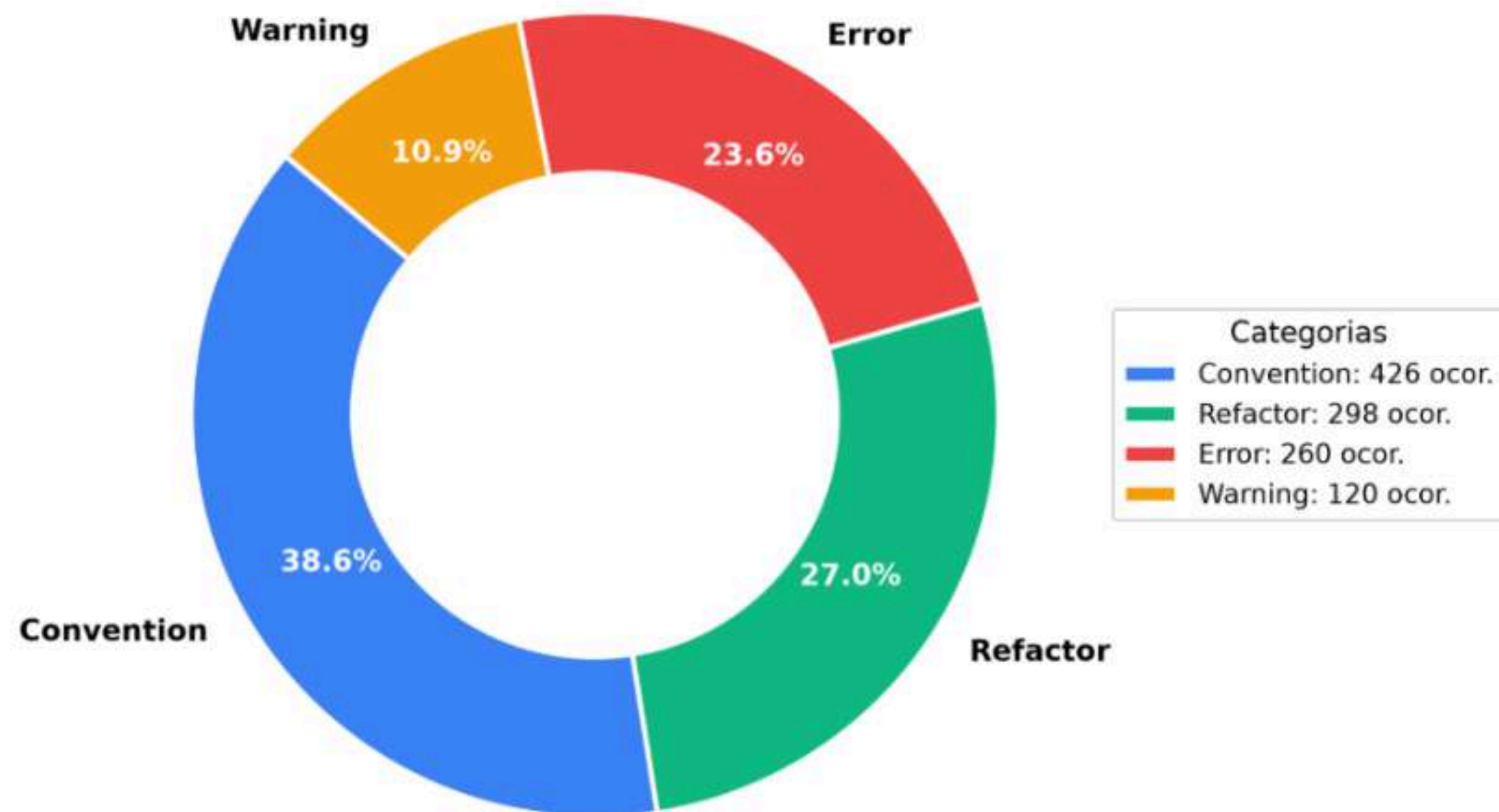
CodeCarbon - Depois da refatoração

duration	emissions	emissions_rate	cpu_energy	ram_energy	energy_consumed
22.93 56486	0.00005177440 3370802725	0.0000022573 769015109355	0.00001687706 215654022	0.00003560395 855556	0.000526440836 3241013

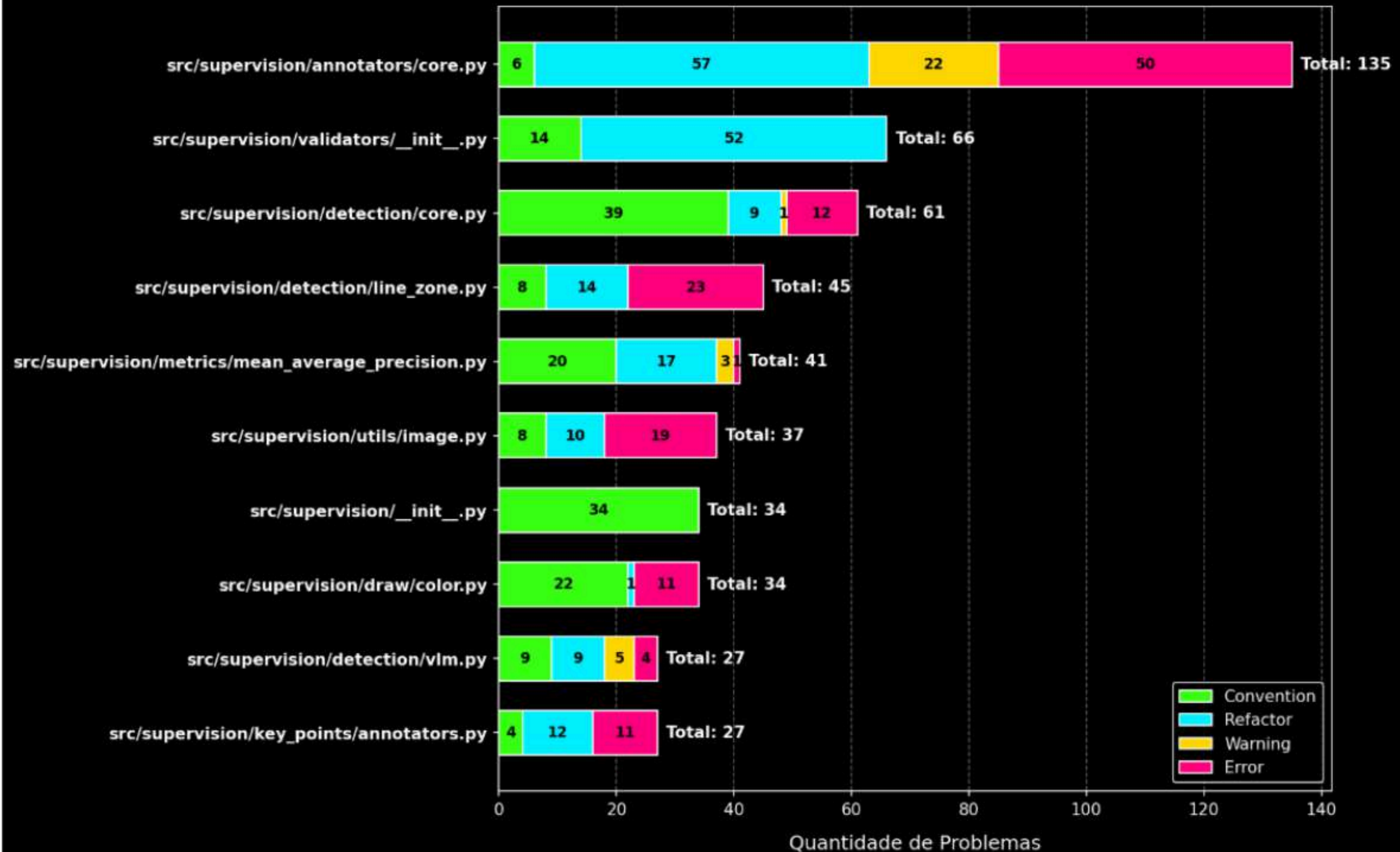
Distribuição de Categorias Pylint Antes da Refatoração



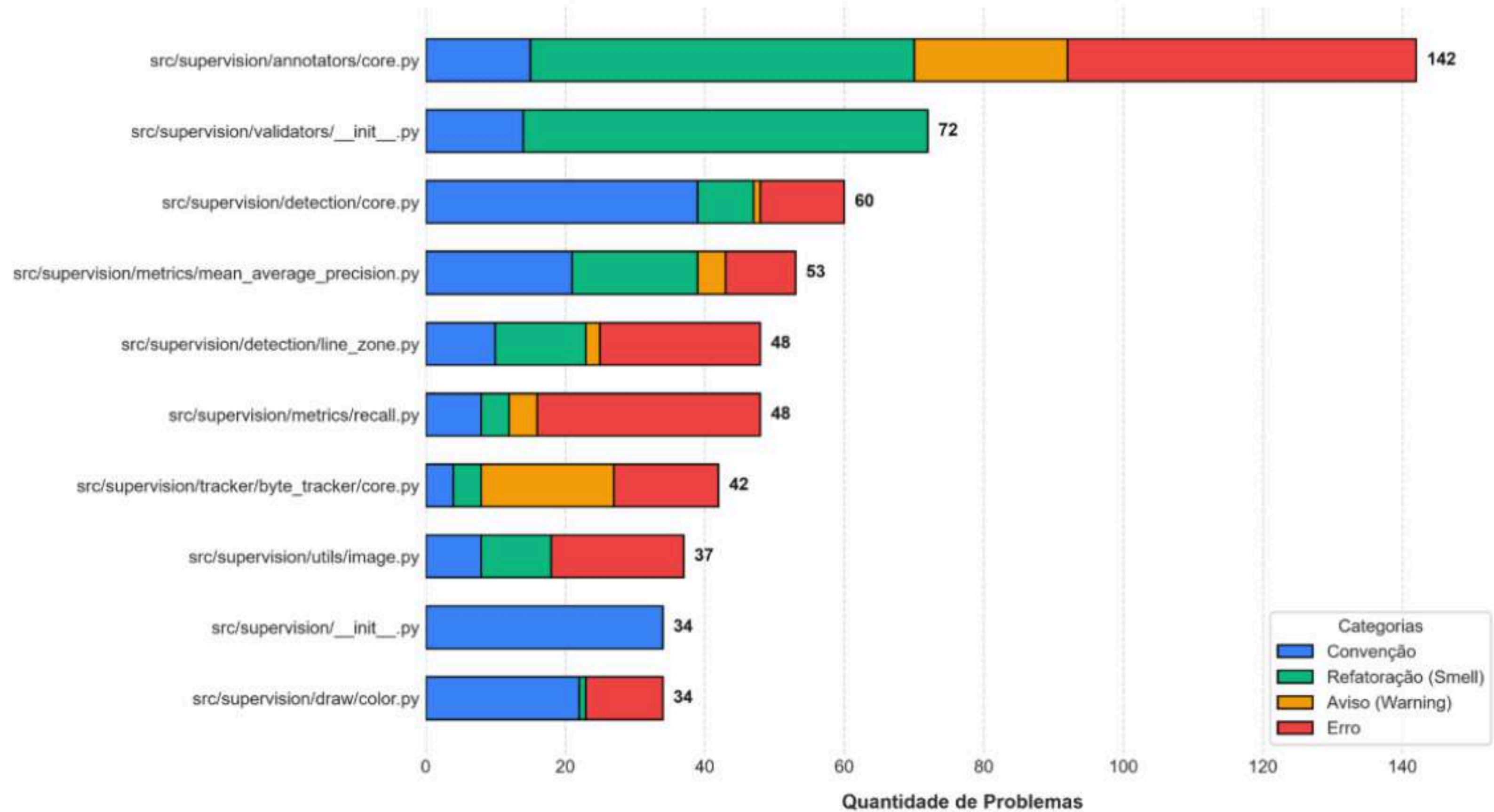
Distribuição das Categorias de Problemas de Código (Pylint)



Top 10 Arquivos mais Críticos Antes da Refatoração (Pylint)



Os 10 Arquivos Mais Críticos (Problemas Pylint)



Obrigado!

Perguntas?